

Forensic Microscopy

Lab 3: Refractive Index

Aritra Bhattacharya
Graduate teaching assistant
ecoevoaritra23@tamu.edu

Where can you find the slides?



(Grab a Microscope from the black closet)

- <https://aritra-evolves.github.io/FIVS215.html>
- Canvas

Today's Agenda

- ❖ Refractive Index - What is it?
 - ❖ Methods of Measuring
 - ❖ Practice
 - ❖ Coming Up
- 

What is refractive index?

In simple terms, it's how much light is bent when it travels through a medium. i.e. glass chips

Effected by: temperature & wavelength

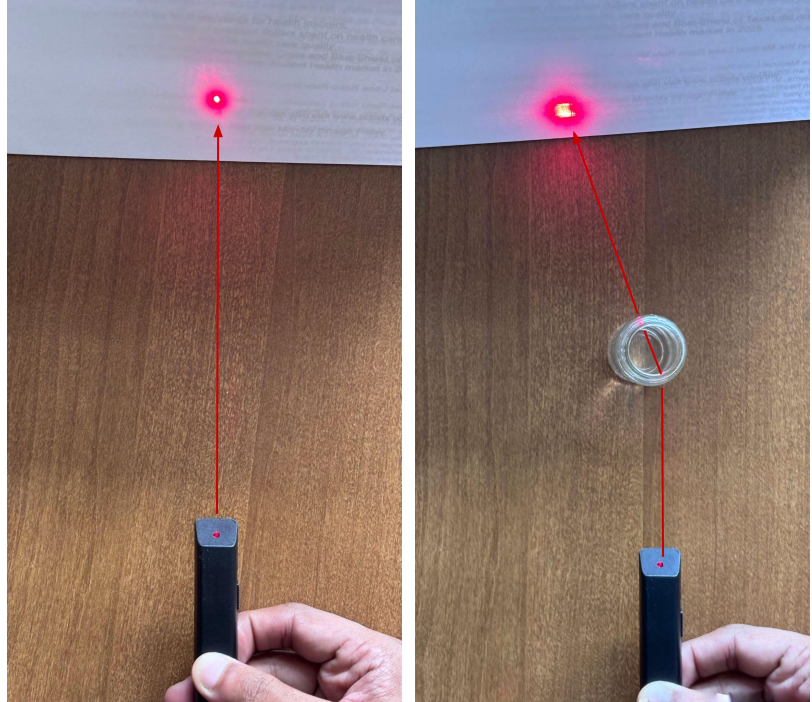
High temp: low refractive index

Low temp: high refractive index

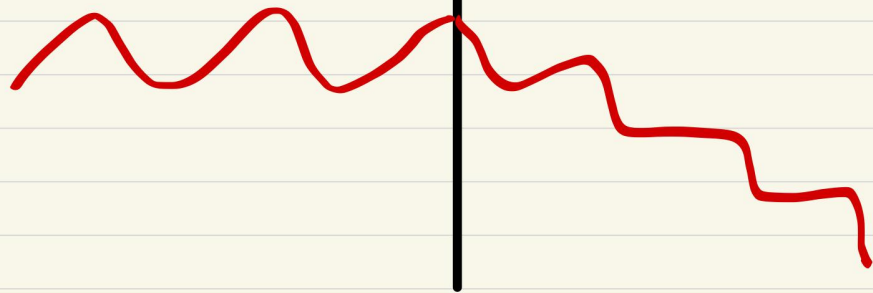
High wavelength: low refractive index

Low wavelength: high refractive index

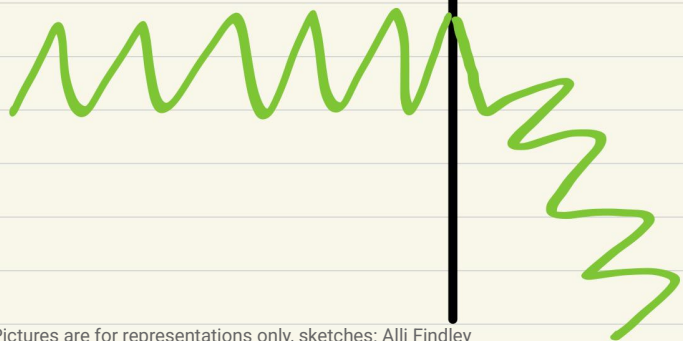
Because of this, we use special filters to focus our light.



long wavelength | low refraction



short wave length | high refraction



Pictures are for representations only, sketches: Alli Findley



Why do we care?

All glass refracts light in one way or another.

- Why we set up Kohler, how eyeglasses work, how hand lens work, etc.

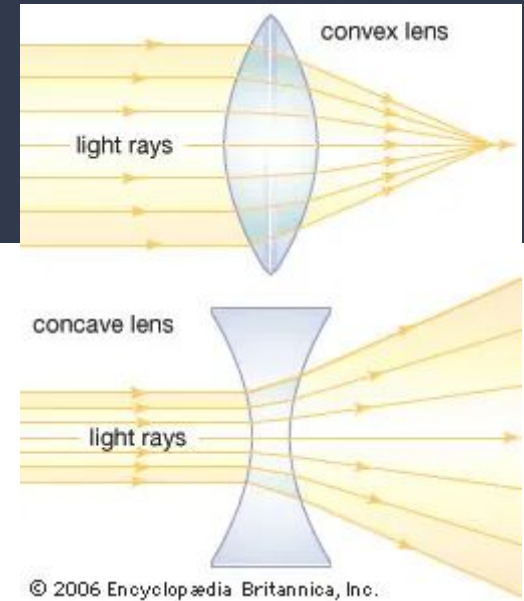
This can be measured and documented for identification purposes.

The way light is bent is consistent for each type of glass, i.e. phone screen glass vs window glass.

Knowing the R.I. can help us rule out or rule in a potential source.

Tells us the class and subclass characteristics of a sample.

- Does NOT tell us if two samples are from the same source



Equations

$$n\lambda = d \sin \theta$$

n = order number

λ = wavelength

d = spacing between the repetitive elements in the structure

$\sin \theta$ = diffraction

This tells us that as wavelength increases diffraction decreases.

You will not need these for LAB. Know them for lecture.

Figure 1 - Refraction of Light

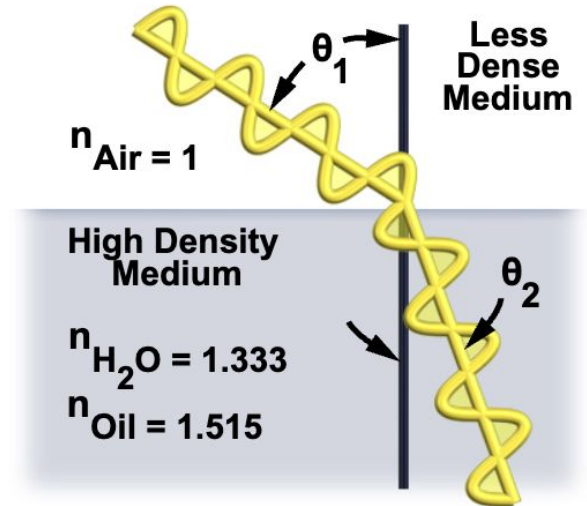


image: <https://www.microscopyu.com/microscopy-basics/refractive-index-index-of-refraction>

Immersion Oils

Placed between the sample and the top slip of the microscope slide.

Has a KNOWN refractive index labelled on the bottle.

Allows to identify the range of a sample's Refractive Index. i.e. is it higher or lower than x

Index is labelled in MM (Meltmount).

Cargille Labs' Meltmount Medium is the gold standard in the field.

N_D indicates the wavelength of light that the oil is best used at – usually a standard of 550nm.



Method 1 – G.R.I.M. Device

Glass Refractive Index Measuring Device

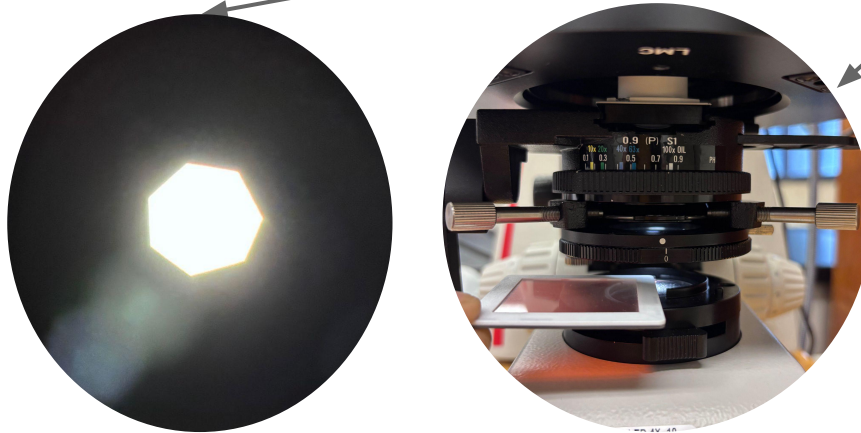
- Computer with a camera
- Automatically measures the R.I. of a sample by heating and cooling the stage.
- Helpful because it does not require the glass chips to be washed after each immersion oil is used.
- We do not have one :(



Foster and Freeman fFTA-1 GRIM3

Method 2 – Becke Lines

- Allows us to test “relative” refractive index.
- This means it tells us the R.I. of the sample RELATIVE to the immersion oil in use.
- Utilizes a “halo” of light around the sample.



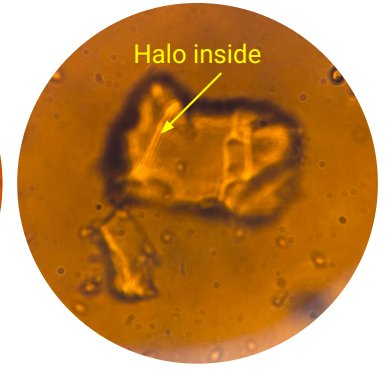
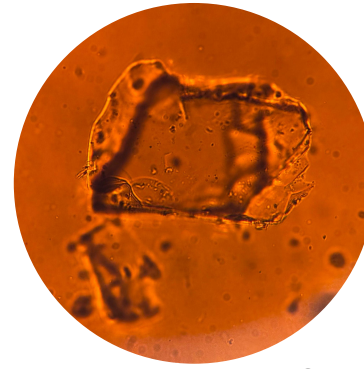
1. Set up Kohler.
2. Close the substage condenser to roughly 20%. This is important.
3. Place a monochromatic filter over the field diaphragm.
4. Observe the halo.
5. “Upward focus” your sample by LOWERING the stage. This increases the working distance between the sample and objective.
6. Observe the direction the Becke line moves.
 - a. The line will move toward the **HIGHER** refractive index.

Becke Lines

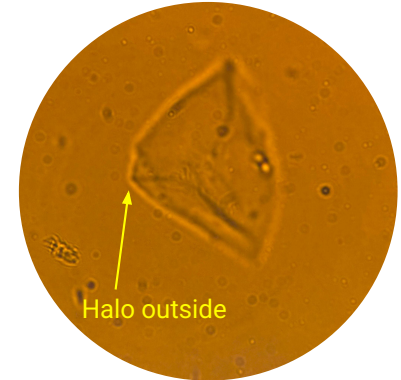
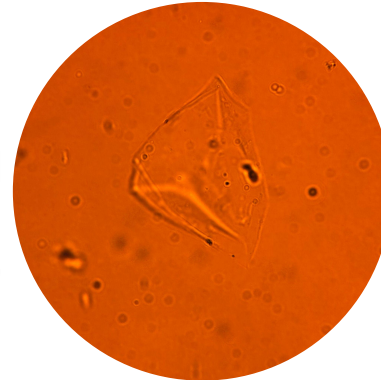
- Rotate the focus knob toward you, to move the stage downward.
- Change the objective lens to higher magnification if the glass chips are too small



Pictures are for representations only



RI of glass is higher



RI of glass is lower

Method 3 – Oblique Lighting

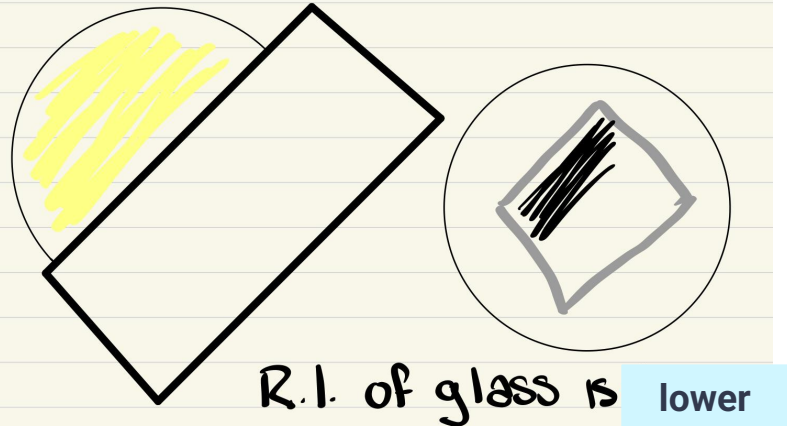
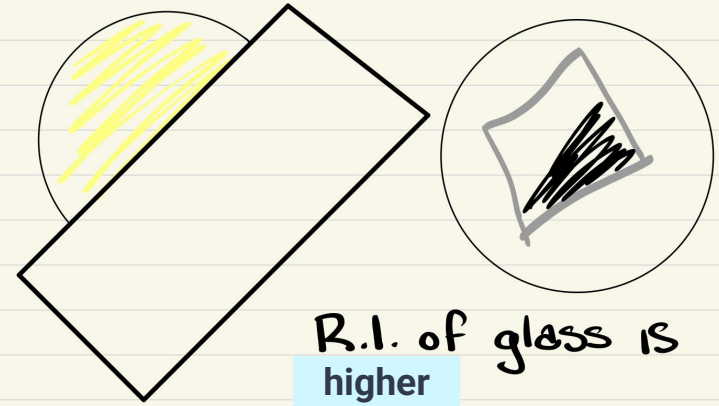
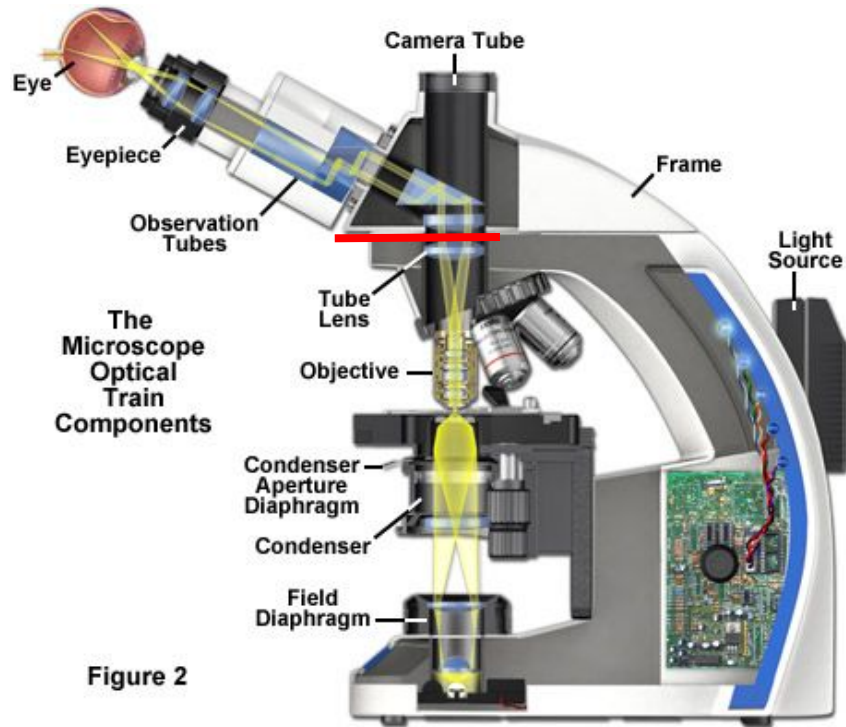
Allows us to test “relative” refractive index.

This means it tells us the R.I. of the sample RELATIVE to the immersion oil in use.

This utilizes shading on the sample.

1. Set up Kohler.
2. Close the substage condenser to roughly 20%. This is important.
3. Place a monochromatic filter over the field diaphragm.
4. Using an index card, cover half the light source over the objective.
5. Observe the shading.
 - a. If the shading is on the SAME side as the covered area, the R.I. of the sample is higher.
 - b. If the shading is on the OPPOSITE side as the covered area, the R.I. of the sample is lower.
 - c. The darker the shading, the greater the difference between the R.I.s

Oblique Lighting



Today's exercise

Check RI of three

Locke glass chip slides (A2, B2, C2)

See you next week!

